Individual Research Gap Analysis

Paper 1: Investigation of Smart Campus Management Platform Based on Digital Twin

Existing Method:

The study creates a Digital Twin-based smart campus management platform by integrating IoT devices, building information, and real-time data. The virtual representation of campus infrastructure is used for monitoring and management.

Advantages:

This approach allows for real-time monitoring, improves how campus facilities are visualized, supports efficient resource management, and gives administrators centralized information about campus operations.

Limitations:

The focus is mainly on monitoring and visualization. Large-scale deployment may need substantial IoT infrastructure, and advanced predictive sustainability analysis is limited.

Research Gap:

The current method does not provide a full solution for combining various sustainability indicators, such as energy consumption, water usage, environmental conditions, and carbon metrics with predictive cloud analytics.

Possible Improvement:

A scalable cloud-based Digital Twin could be developed to combine multiple sustainability parameters and use predictive analytics to identify unusual consumption and predict future resource needs.

Paper 2: Developing Campus Digital Twin Using Interactive Visual Analytics Approach

Existing Method:

The research creates a campus Digital Twin that uses interactive visual analytics. This combines spatial and operational campus information and gives users an interactive view of campus conditions.

Advantages:

This approach improves data visualization. It allows users to explore complex campus information more effectively, helping them with planning and decision-making.

Limitations:

The work focuses more on visualization and interaction than on automated sustainability prediction and optimization.

Research Gap:

There is an opportunity to link visual Digital Twins with real-time cloud analytics. This connection can continuously assess campus sustainability performance.

Possible Improvement:

Combine cloud-based data processing, sustainability indicators, anomaly detection, and predictive models with the interactive Digital Twin.

Paper 3: A Smart Campus' Digital Twin for Sustainable Comfort Monitoring

Existing Method:

The study uses Digital Twin technology and environmental sensing to monitor conditions that affect occupant comfort in a smart campus.

Advantages:

It allows for continuous environmental monitoring and helps to clarify the relationship between building conditions, comfort, and resource use.

Limitations:

The system focuses mainly on comfort and environmental conditions, rather than offering a complete campus sustainability management solution.

Research Gap:

Energy, water, carbon emissions, waste, and other sustainability indicators are not fully combined in one campus-level analytics platform.

Possible Improvement:

Expand the Digital Twin to include multiple campus buildings and merge energy, water, occupancy, and environmental data into a single sustainability dashboard.

Paper 4: Multifunctional Models in Digital and Physical Twinning of the Built Environment—A University Campus Case Study

Existing Method:

The research looks at university campus infrastructure in two ways: physical. It uses Digital Twin. Building modelling concepts to do this.

Advantages:

This method gives us a really good idea of what the physical assets are and it helps us understand and manage the built environment better.

Limitations:

Making and keeping these detailed models up to date can be very hard on computers and it is also hard to keep them in sync with what is actually happening.

Research Gap:

Not many people are using native analytics to turn Digital Twin information into suggestions for how to be more sustainable.

Improvement:

We can connect the campus model to cloud services and real-time sensor streams. This will help us calculate how sustainable things are and give us ideas for what we can do to improve things. We can use Digital Twin and cloud services to make the campus more sustainable. The Digital Twin will help us make decisions, about the campus.

Paper 5: BIM-Based Digital Twin Development for University Campus Management: Case Study ETSICCP

Existing Method:

The study integrates BIM with Digital Twin concepts to digitally represent university facilities and support campus management.

Advantages:

It provides detailed asset information, improves visualization, and supports maintenance and infrastructure management.

Limitations:

BIM-based representations can require significant effort to maintain, particularly when physical campus conditions change.

Research Gap:

The approach has limited integration of predictive sustainability analytics and automated cloud-based processing.

Possible Improvement:

Combine BIM with IoT sensors and cloud services so that the Digital Twin is automatically updated and sustainability trends can be predicted.

Paper 6: The Development of a 3D Model Based on Digital Twin for Campus Master Plan Optimization

Existing Method:

A three-dimensional Digital Twin model is used to represent campus infrastructure and support master planning and spatial optimization.

Advantages:

The model improves visualization, infrastructure planning and evaluation of campus development scenarios.

Limitations:

The primary emphasis is on planning and spatial representation, than continuous operational sustainability monitoring.

Research Gap:

Real-time resource consumption and environmental data are not sufficiently incorporated into planning decisions.

Possible Improvement:

Integrate cloud analytics with the 3D model so that planning decisions can also consider energy, occupancy and environmental performance.

Paper 7: Digital Twins for Smart Campus Networks: An End-to-End Framework for Multi-Domain Data Intelligence

Existing Method:

The study suggests a Digital Twin framework that brings together information from smart-campus areas to allow combined data intelligence.

Advantages:

It helps with analysis across areas, central monitoring and better understanding of connected campus systems.

Limitations:

Combining systems makes the design more complicated and causes problems with how systems work together and with security.

Research Gap:

Information from areas, in a Digital Twin could be used much more for sustainability scoring predicting future trends and making improvements.

Possible Improvement:

Add cloud data pipelines and sustainability analysis tools that bring together energy data, environmental data, people movement and building structures.

Paper 8: Digital Twins: Cornerstone to Circular Economy and Sustainability Goals

Existing Method:

The study looks at how Digital Twin technology can help with sustainability goals and circular economy goals by making it easier to watch resources and make decisions about their life cycles.

Advantages:

It offers a view of sustainability and finds many chances to use Digital Twin technology for better use of resources.

Limitations:

The research is mostly theoretical and based on reviews. Does not show a full smart-campus setup.

Research Gap:

There is a need to take the ideas about Digital Twin and sustainability and turn them into real working applications, on a campus.

Possible Improvement:

Add sustainability measures inside a campus Digital Twin and use cloud analytics to track and compare how well each building is doing.

Paper 9: Linking Digital Twin Paradigm for Urban Heat Monitoring and Policy Integration to Building Smart City Climate Resilience

Existing Method:
Digital Twin concepts are applied to environmental and urban heat information to support climate-resilience planning.

Advantages:
The approach combines environmental monitoring with decision support and demonstrates the usefulness of Digital Twins for sustainability planning.

Limitations:
It targets urban-scale climate resilience rather than university campus operations.

Research Gap:
Campus-level environmental conditions could be combined with resource-consumption data to provide more localized sustainability decisions.

Possible Improvement:
Adapt environmental Digital Twin analytics to campus buildings and combine temperature and weather information with energy and occupancy data.

Paper 10: A BIM-Integrated Eco-Digital Framework for Predictive Maintenance and Sustainability Assessment in Educational Buildings Towards Digital Twin Readiness

Existing Method:

The framework uses methods to make campus energy systems, renewable resources, microgrids and resilience better.

Advantages:

It makes energy management better helps with adding renewable energy sources and looks at resilience.

Limitations:

Its focus on sustainability is mostly on energy.

Research Gap:

parts of campus sustainability like water how many people are, in buildings, air quality and carbon levels need to be studied together.

Possible Improvement:

Link energy optimization with a Digital Twin that includes parts of sustainability and give a single cloud dashboard.

Paper 11: Sustainable Energy Transitions in Smart Campuses: An AI-Driven Framework Integrating Microgrid Optimization, Disaster Resilience, and Educational Empowerment

Existing Method:

The study proposes an AI-driven framework for sustainable energy management in smart campuses by integrating microgrid optimization, renewable energy resources, disaster resilience, and intelligent decision-making. It focuses on improving energy efficiency and ensuring reliable campus power management.

Advantages:

The framework enhances energy utilization, supports renewable energy integration, improves disaster preparedness, and promotes intelligent energy management through AI-based optimization techniques.

Limitations:

The proposed framework primarily focuses on campus energy systems and microgrid optimization. It does not provide a comprehensive Digital Twin platform that integrates multiple sustainability indicators such as water consumption, occupancy, indoor environmental conditions, and real-time visualization.

Research Gap:

The study lacks a unified cloud-based Digital Twin capable of combining energy management with other sustainability parameters, predictive analytics, real-time IoT monitoring, and interactive visualization for complete campus sustainability assessment.

Possible Improvement:

Develop a cloud-based Digital Twin platform that integrates AI-driven energy optimization with IoT sensor data, water usage, occupancy, environmental monitoring, sustainability scoring, predictive analytics, and interactive dashboards using AWS cloud services.

Paper 12: From Building Information Modelling to Digital Twins: Digital Representation for a Circular Economy

Existing Method:

The research examines the transition from static BIM representations toward dynamic Digital Twins for lifecycle and circular-economy applications.

Advantages:

It improves lifecycle information management and demonstrates how Digital Twins can support sustainable asset management.

Limitations:

The approach does not primarily address real-time campus analytics and cloud implementation.

Research Gap:

A mechanism is required to connect digital building representations continuously with operational sustainability data.

Possible Improvement:

Use IoT data pipelines and cloud services to synchronize physical campus resources with their Digital Twin representations.

Paper 13: Digital Twin-Enabled Smart Facility Management: A Bibliometric Review

Existing Method:

The study performs bibliometric analysis of existing research concerning Digital Twins and smart facility management.

Advantages:

It identifies major research themes, technological trends, and emerging directions.

Limitations:

As a review, it does not experimentally implement or evaluate a Digital Twin platform.

Research Gap:

Practical implementations integrating facility management, sustainability analytics, and cloud infrastructure remain necessary.

Possible Improvement:

Build and evaluate a working cloud-based campus Digital Twin using real or simulated operational data.

Paper 14: A Review of Building Digital Twins to Improve Energy Efficiency

Existing Method:

The study looks at how Digital Twin techniques used to keep an eye on and improve the energy performance of buildings.

Advantages:

It talks about the ways to optimize energy use and shows how Digital Twins can help reduce the amount of energy buildings use.

Limitations:

The study mainly focuses on one building at a time and how to make it more energy efficient.

Research Gap:

When it comes to making a whole campus sustainable we need to look at more than one building and energy use. We need to consider multiple buildings and resources.

Possible Improvement:

We can make things better by creating a Digital Twin for the campus that brings together information from many buildings and looks at sustainability in a bigger way all, in one online platform.

Paper 15: AI-Driven Digital Twins for Enhancing Indoor Environmental Quality and Energy Efficiency in Smart Building Systems

Existing Method:
The study combines AI and Digital Twin technology to analyze indoor environmental conditions and improve energy efficiency in smart buildings.

Advantages:
It supports intelligent energy optimization, environmental monitoring, fault identification, and improved occupant comfort.

Limitations:
The approach is primarily building-centric and focuses strongly on indoor environmental quality and energy.

Research Gap:
There is scope for extending AI-enabled Digital Twins from individual smart buildings to integrated campus-level sustainability management.

Possible Improvement:
Create a cloud-based multi-building Digital Twin where AI models analyze energy, environmental, occupancy, and other sustainability parameters together.

Project Objectives

- To develop a cloud-based Digital Twin platform that creates a virtual representation of multiple campus buildings and enables centralized monitoring of campus operations.
- To integrate multiple sustainability parameters such as energy consumption, water usage, occupancy, temperature, humidity, and estimated carbon emissions into a unified Digital Twin environment.
- To develop a dynamic Campus Sustainability Score for evaluating and comparing the sustainability performance of campus buildings based on resource consumption and environmental indicators.
- To implement predictive analytics for forecasting energy and water consumption and identifying abnormal resource usage or potential wastage.
- To develop a what-if simulation and analytics dashboard that evaluates different resource-management scenarios and visualizes sustainability scores, predictions, alerts, and recommendations to support campus decision-making.